AI Edge (Computing)

1. What is AI and ML?

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1

Human Intelligence vs Artificial Intelligence



- Human Intelligence
  - Solve problems
  - Achieve goals
  - Analyse & reason
  - Communicate, collaborate & influence
  - Consciousness, emotions, intuition, imagination

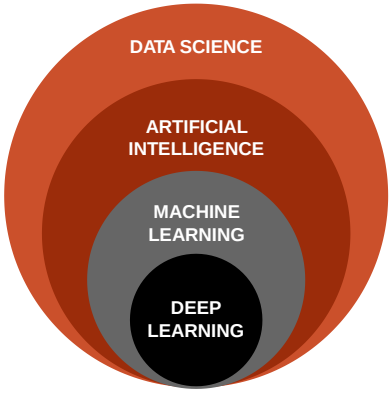


- Artificial Intelligence
  - The ability for machines to simulate & enhance (human) intelligence

2

Machine Learning (ML) is a subfield of AI

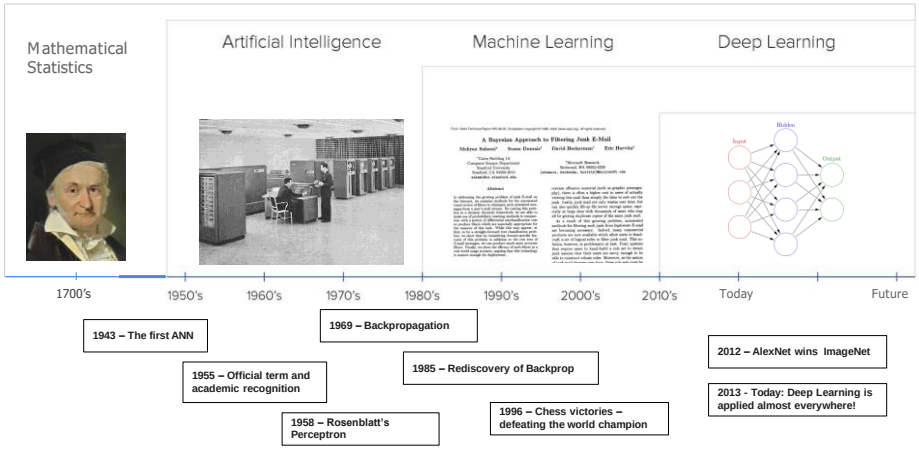
And Deep Learning is a subfield of ML



3

AI is not new

It has been around for a long time!



4

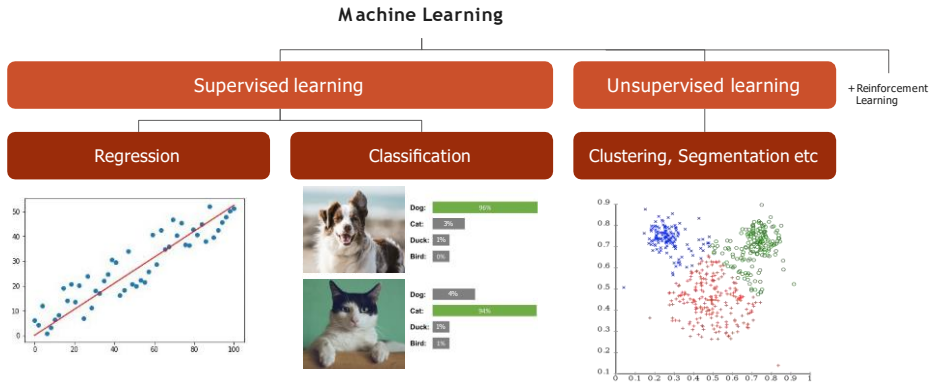
What is different this time?

4 key enablers

1. Data availability
2. Computational power
3. Algorithm advancements
4. Broad public interest

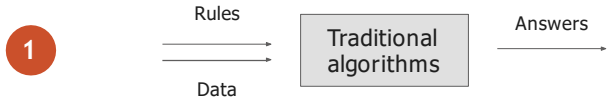
5

The categories of Machine Learning



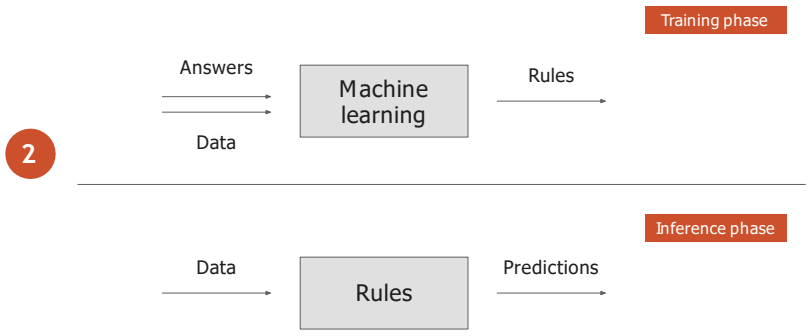
6

Traditional algorithms rely on rules defined by humans



7

Machine Learning algorithms extract rules from data



8

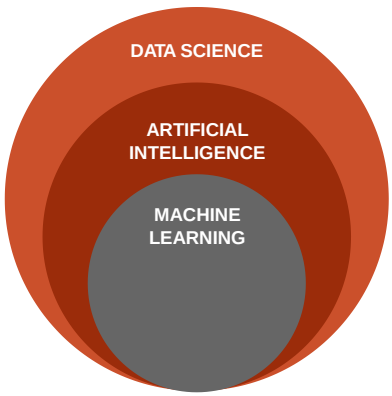
What is (tiny) Machine Learning?

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9

What is Machine Learning?

1. Machine learning is a subfield of Artificial Intelligence focused on developing algorithms that learn to solve problems by analyzing data for patterns

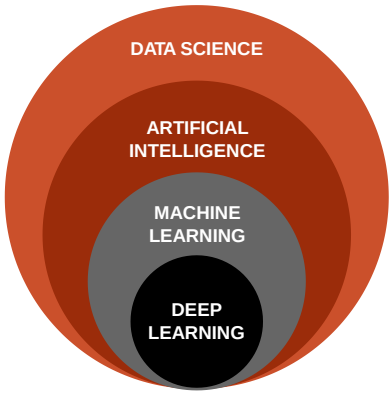


10

What is Deep (Machine) Learning?

1. Machine learning is a subfield of Artificial Intelligence focused on developing algorithms that learn to solve problems by analyzing data for patterns

2. Deep Learning is a type of Machine Learning that leverages Neural Networks and Big Data



11

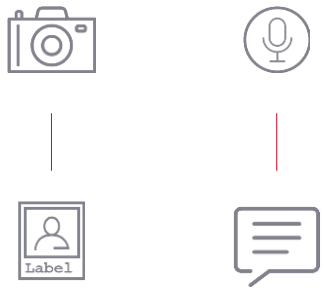
Applications of Machine Learning





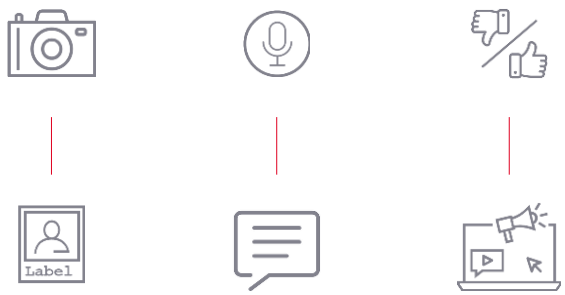
12

— Applications of Machine Learning



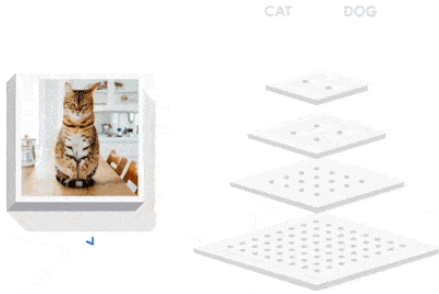
13

— Applications of Machine Learning



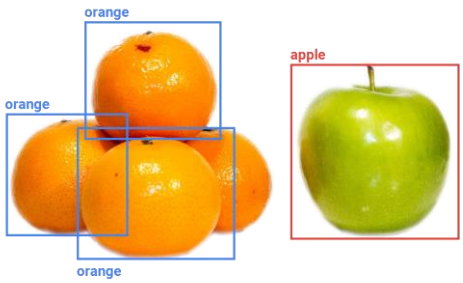
14

Image Classification



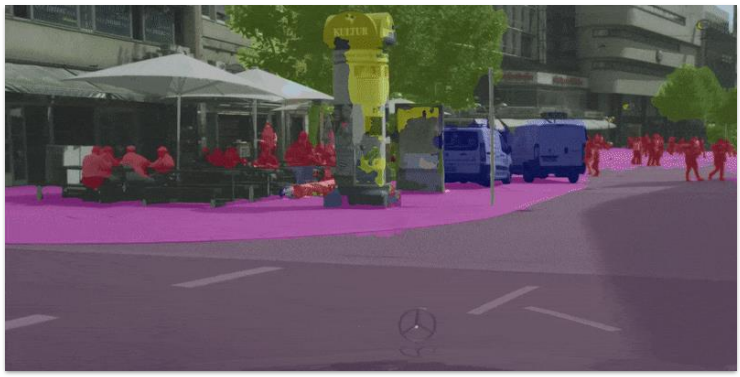
15

Object Detection



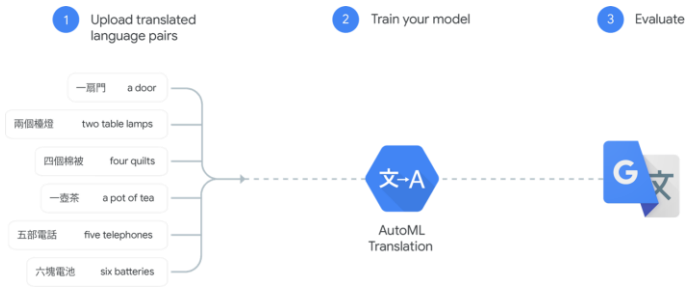
16

— Segmentation



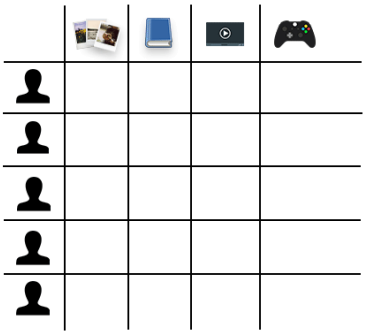
17

— Machine Translation



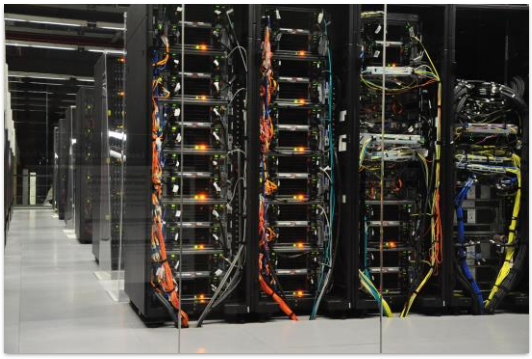
18

Recommendations



19

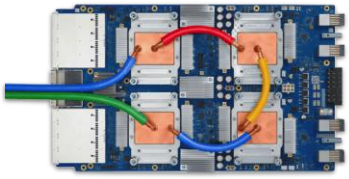
Datacenter





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— TPUs & GPUs





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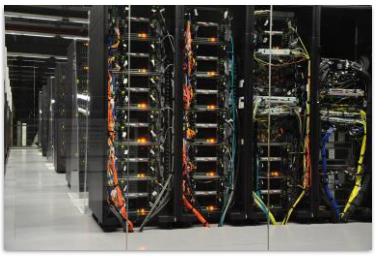
— Processing power



Bigger Is Not Always Better.

22

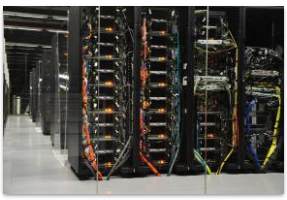
From datacenter to edge





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From datacenter to edge



High power
High bandwidth
High latency



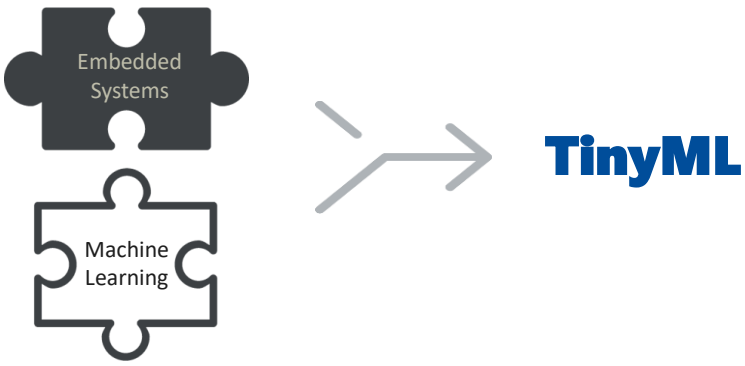
Low power
Low bandwidth
Low latency

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How do we enable TinyML?

25

What makes TinyML?



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What makes TinyML?

Let's take an example





Google Assistant

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What makes TinyML?

Let's take an example





Hi, how can I help?

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What makes TinyML?

The three basic steps

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What makes TinyML?

The three basic steps



Step 1
Audio input
from microphone (sensor)

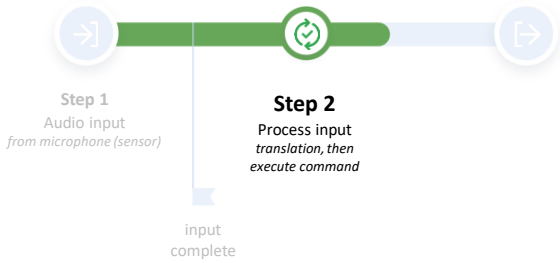
input complete

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What makes TinyML?

The three basic steps



How do we enable TinyML

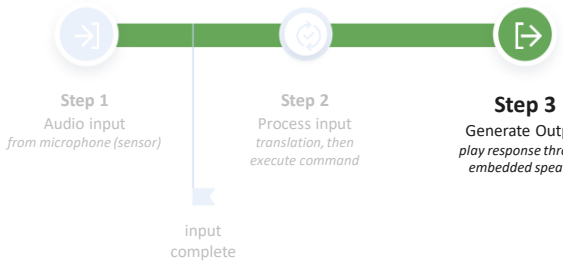
31

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31

What makes TinyML?

The three basic steps



How do we enable TinyML

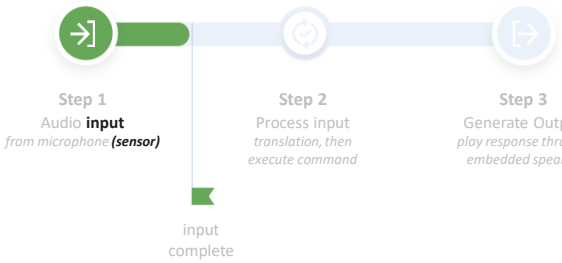
32

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32

Input

The three basic steps



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Endpoints have sensors, tons of sensors

An overview

Motion Sensors Gyroscope, radar, magnetometer, accelerator	**Acoustic Sensors** Ultrasonic, Microphones, Geophones, Vibrometers	**Environmental Sensors** Temperature, Humidity, Pressure, IR, etc.
Touchscreen Sensors Capacitive, IR	**Image Sensors** Thermal, Image	**Biometric Sensors** Fingerprint, Heart rate, etc.
Force Sensors Pressure, Strain	**Rotation Sensors** Encoders	

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Endpoints have sensors, tons of sensors

An overview

Motion Sensors Gyroscope, radar, magnetometer, accelerator	**Acoustic Sensors** Ultrasonic, Microphones, Geophones, Vibrometers	**Environmental Sensors** Temperature, Humidity, Pressure, IR, etc.
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Force Sensors Pressure, Strain	**Rotation Sensors** Encoders	

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Biometric sensors



Non-invasive Glucose Monitoring



Fingerprint + Photoplethysmography (**PPG**)

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Endpoints have sensors, tons of sensors

An overview

Motion Sensors
Gyroscope, radar, magnetometer, accelerator

Acoustic Sensors
Ultrasonic, Microphones, Geophones, Vibrometers

Environmental Sensors
Temperature, Humidity, Pressure, IR, etc.

Touchscreen Sensors
Capacitive, IR

Image Sensors
Thermal, Image

Biometric Sensors
Fingerprint, Heart rate, etc.

Force Sensors
Pressure, Strain

Rotation Sensors
Encoders

37

Endpoints have sensors, tons of sensors

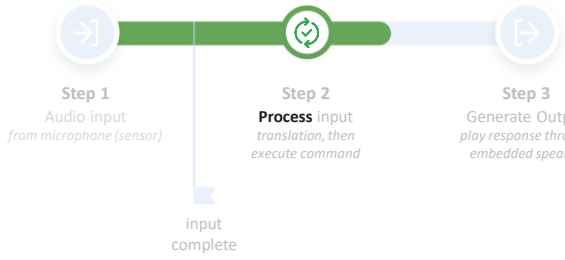
An overview



38

Processing

The three basic steps



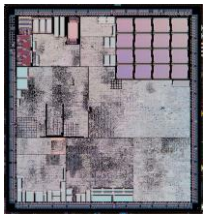
39

Thinking BIG



40

— Thinking BIG



41

— Thinking BIG

BIG
GPU / CPU
561mm²

42

— Thinking SMALL

BIG
GPU / CPU
561mm²



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43

— Thinking SMALL

BIG
GPU / CPU
561mm²



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— Thinking SMALL

BIG
GPU / CPU
561mm²

SMALL
Mobile SoC
83mm²

45

— Thinking TINY

BIG
GPU / CPU
561mm²

SMALL
Mobile SoC
83mm²



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— Thinking TINY

BIG
GPU / CPU
561mm²

SMALL
Mobile SoC
83mm²

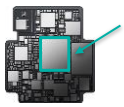


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— Thinking TINY

BIG
GPU / CPU
561mm²

SMALL
Mobile SoC
83mm²



48

Thinking TINY





Mobile SoC
83mm²

Apple 0778
30mm²

How do we enable TinyML

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49

Thinking Record-breaking





Mobile SoC
83mm²

Apple 0778
30mm²

Kinetis KL03
3.2mm²

How do we enable TinyML

50

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50

— Thinking Record-breaking

BIG
GPU / CPU
561mm²

SMALL
Mobile SoC
83mm²

TINY
Apple 0778
30mm²

Kinetis KL03
3.2mm²

world's smallest ARM-Powered MCU
48MHz, 32KB flash, 20-pin

How do we enable TinyML

51

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51



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MCU Demand & Pricing Forecast

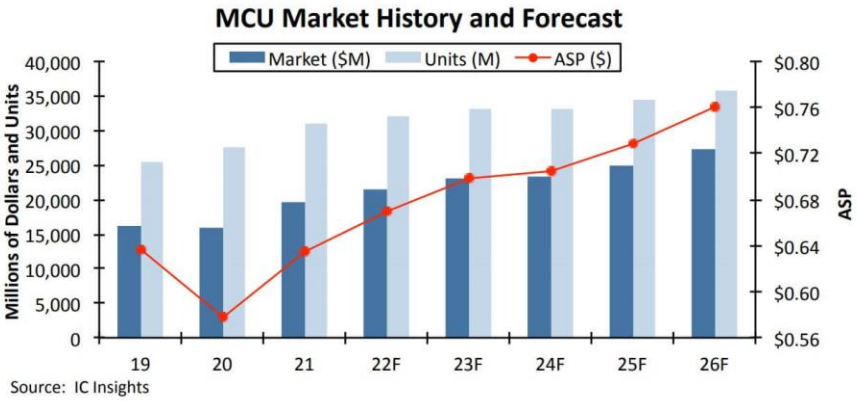
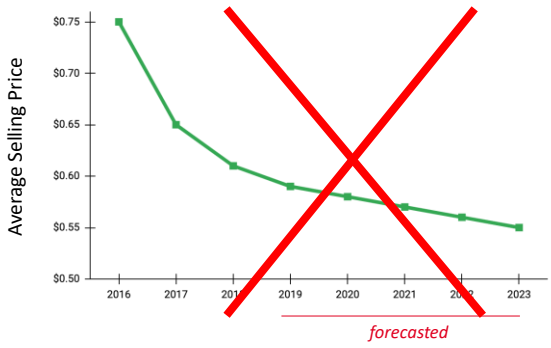


Figure 1

53

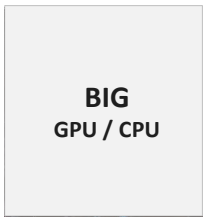
MCU Pricing Forecast



Source: IC Insights

54

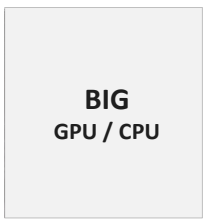
Comparing power



300W
NVIDIA Tesla K80

55

Comparing power



300W
NVIDIA Tesla K80



3.64W
Apple A12

56

Comparing power

BIG
GPU / CPU

300W
NVIDIA Tesla K80

SMALL

3.64W
Apple A12

140 µW
Syntiant NDP100

How do we enable TinyML

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57

Comparing power

BIG
GPU / CPU

300W
NVIDIA Tesla K80

SMALL

3.64W
Apple A12

Neural Decision Processor
Always-on deep learning speech/audio recognition
Ultra low power, 128KB SRAM, 12-pin, 2.52mm²

140 µW
Syntiant NDP100

How do we enable TinyML

58

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58

Comparing power



Use case: button cell battery

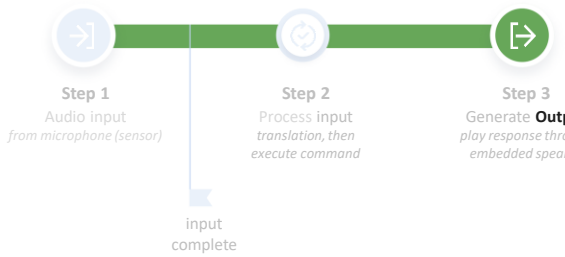
Neural Decision Processor
Always-on deep learning speech/audio recognition
Ultra low power, 128KB SRAM, 12-pin, 2.52mm²

140 µW
Syntiant NDP100

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Output

The three basic steps



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— Output

Physical outputs





Digital outputs



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— MCUs enable TinyML

SIZE

LOW POWER

LOW COST

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MCUs enable TinyML

SIZE | LOW POWER | LOW COST



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63

MCUs enable TinyML

SIZE | LOW POWER | LOW COST



< 140 µW
Syntiant NDP100

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64

MCUs enable TinyML

SIZE — LOW POWER — LOW COST

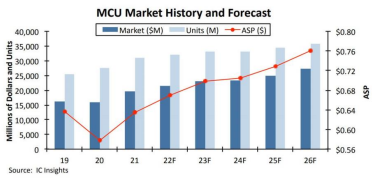


Figure 1

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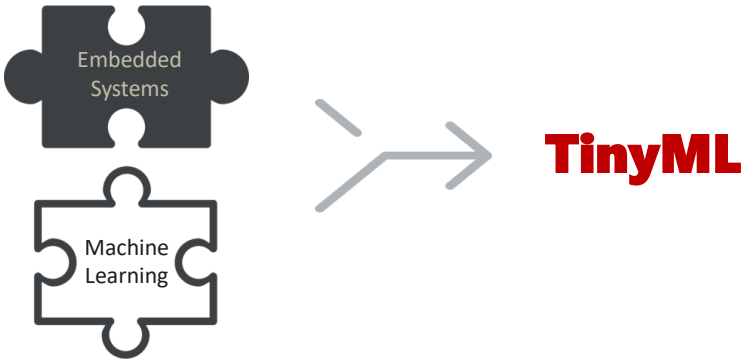
MCUs enable TinyML

SIZE — LOW POWER — LOW COST

HIGH DEMAND

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What makes TinyML?



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Some questions

Artificial Intelligence (AI) is only concerned with the designing and building of intelligent agents that receive precepts from the environment and discover patterns in that environment.

a) True

b) False

68

Some questions

A classification model trained with data that has ground truth labels is what kind of machine learning?

a) Regression
b) Supervised Learning
c) Unsupervised Learning
d) Reinforcement Learning

70

Some questions

Statement: All deep learning is machine learning.

a) True
b) False

72

Some questions

Why is there a bright future for machine learning on embedded devices?

a) Smaller machine learning models are more accurate
b) Deterministic programming is sufficient for solving all embedded problems
c) Emerging problems require low-power devices
d) All of the above

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Limitations and Ethics

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The flipside of the 4 key AI enablers…

Do you remember them?

1	2	3	4
Data availability	Computational power	Algorithm advancements	Broad public interest
Violate privacy & data integrity	**Energy & capital intensive**	**Introduction of biases & opacity**	**Hype vs reality**

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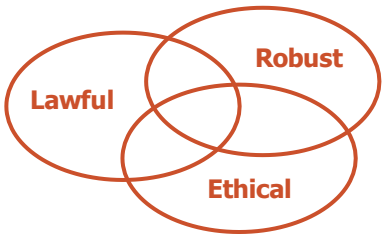
Trust in …

Trust in Humans

Selection of components

- Morals & Ethics
- Character
- Societal Laws
- Cultural Laws
- Compassion
- …

Trust in AI



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7 key requirements for Trustworthy AI

1. Human agency & oversight
2. Technical robustness & safety
3. Privacy & data governance
4. Transparency
5. Diversity, fairness & non-discrimination
6. Societal & environmental wellbeing
7. Accountability



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Responsible AI: Human-Centered Design

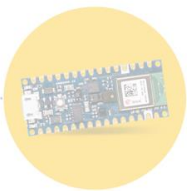


Course 1
Fundamentals of TinyML

- **What am I building?**
- Who am I building this for?
- What are the consequences for the user if it fails?

Course 2
Applications of TinyML

Course 3
Deploying TinyML



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What am I building

Traditional Programming

Machine Learning



Which one **should** we choose?

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Traditional Programming

Pros

- Quicker to build
- Easier to explain
- Easier to debug
- Easier to maintain
- More consistent/stable



```
if (speed < 4) {
   status = WALKING;
}
```

Cons

- Does not scale
- Does not adapt to changes
- Does not work for complex tasks

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— Machine Learning

Pros

- Complex problems
- Scale
- Adaptable
- Personalization
- Improves over time



```
01010010101100101010
10010101010010111101
01001010100101010001
01010010101001010010
```

Label = WALKING

Cons

- Slower to build
- Harder to explain/interpret
- Harder to debug

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— What is the goal?

1 — **What** is the problem?

2 — **Why** does the problem need to be solved?

3 — **How** could the problem be solved manually?

4 — Will AI offer **unique** value?

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Design AI for social good



There are **17 goals** on the United Nations' 2030 Agenda for Sustainable development, for example:

- Ending poverty and world hunger
- Improving health and education
- Reducing inequality and injustice

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Challenges facing AI4SG



- Learning from **limited data**
- Geographical **imbalances**
- **Biased** data
- Learning with **limited** memory and computation

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Window of opportunity with TinyML

- Learning with limited memory
- and computation
- Battery-operated
- On-device computing
- Low latency
- Low cost
- Small size



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Some examples where TinyML is used

Predictive maintenance and monitoring of remote wind turbines

→ Challenging, time-consuming

→ Cost savings, reduced down times, better availability for higher reliability, higher quality of service

Image: Ping Services, magnetic IoT device that analyses sensor data on the edge and reports to the cloud



88

Some examples where TinyML is used

Analysis of the cassava crop using an offline phone app (Nuru)

→ Identify deseases and treat them

→ Real-time mitigation without internet

Next step: Leveraging TinyML and Tensorflow Lite for Microcontrollers to deploy sensors across remote farms for better analysis and tracking.



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Some examples where TinyML is used

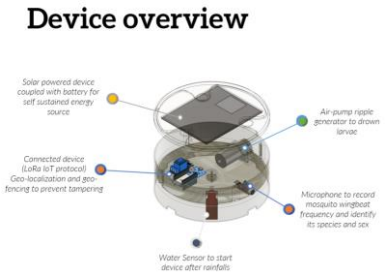
Solar Scare Mosquito:

→ Sensor to decrease the spread of mosquito-born epidemics (Malaria, Dengue, Zika)

→ Disruption of breeding cycle by agitating water

Works on solar power, uses rain and acoustic sensors to conserve battery.

Sends over data over low power low-speed communication protocol.



90

Some examples where TinyML is used

Ecological and environmental monitoring:

Early warning system using acoustic and thermal sensors in a custom machine learning model, using solar power

→ Warning system for train drivers

→ Minimize fatal collisions with elephants



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Some examples where TinyML is used

Preventing whales from hitting ships in busy shipping lanes

→ Constant real-time monitoring

→ Increased density of sensor deployments improves system efficiency and efficacy



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Importance of AI

Human-centered, responsible

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Responsible AI: Human-Centered Design

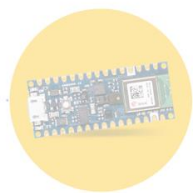
START | DESIGN | DEVELOPMENT | DEPLOYMENT | END

Course 1
Fundamentals of TinyML

- What am I building?
- Who am I building this for?
- **What are the consequences for the user if it fails?**

Course 2
Applications of TinyML

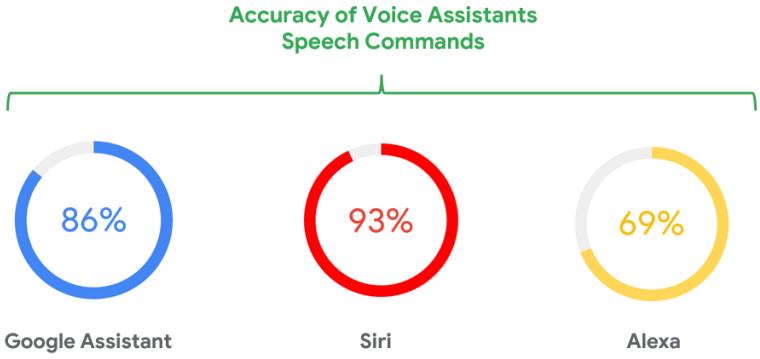
Course 3
Deploying TinyML



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AI isn't perfect



Source: https://loupventures.com/annual-digital-assistant-iq-test/

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When AI succeeds

The confusion matrix

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	**True Positive**	False Positive
Predicted Disease = No	False Negative	**True Negative**

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When AI fails

The confusion matrix

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	True Positive	**False Positive**
Predicted Disease = No	**False Negative**	True Negative

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Two types of error

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	True Positive	False Positive ***Type 1 Error***
Predicted Disease = No	False Negative ***Type 2 Error***	True Negative

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Two types of error = two ways to optimize

Choose:

- High precision
  - High accuracy of positive predictions
- High recall
  - High completeness of positive predictions

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Precision

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	True Positive	False Positive *Type 1 Error*
Predicted Disease = No	False Negative *Type 2 Error*	True Negative

If we optimize our model for **high precision**



We **minimize** *Type 1* Errors
(fewer false positives)

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Precision

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	True Positive	False Positive *Type 1 Error*
Predicted Disease = No	False Negative *Type 2 Error*	True Negative

If we optimize our model for **high precision**



We **minimize** *Type 1* Errors
(fewer false positives)



We **increase** *Type 2* Errors
(more false negatives)

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Recall

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	True Positive	False Positive *Type 1 Error*
Predicted Disease = No	False Negative *Type 2 Error*	True Negative

If we optimize our model for **high recall**



We **minimize** *Type 2* Errors
(fewer false negatives)

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— Recall

	Actual Disease = Yes	Actual Disease = No
Predicted Disease = Yes	True Positive	False Positive *Type 1 Error*
Predicted Disease = No	False Negative *Type 2 Error*	True Negative

If we optimize our model for **high recall**



We **minimize** *Type 2* Errors
(fewer false negatives)

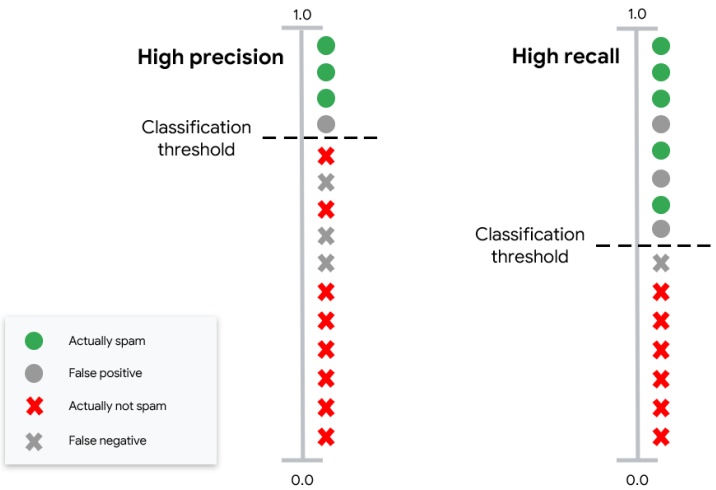


We **increase** *Type 1* Errors
(more false positives)

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— Tension between Precision and Recall



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Which one is best or more important? It depends!

Ask yourself 4 questions

1. Who are the key stakeholders (both direct and indirect)?
2. What are the consequences of a ***Type 1*** error?
3. What are the consequences of a ***Type 2*** error?
4. Which type of error would be most harmful?

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Which one is best or more important? It depends!

Ask yourself 4 questions

4. Which type of error would be most harmful?

If ***Type 1*** errors are most harmful,
then **precision** is more important

If ***Type 2*** errors are most harmful,
then **recall** is more important

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Which one is best or more important? It depends!

Examples

Spam Filtering



1	Key stakeholders?	user with email account; user's correspondents
2	***Type 1*** error?	user doesn't receive important email, correspondent is unable to communicate with user
3	***Type 2*** error?	user sees spam in email inbox
4	Most harmful?	***Type 1*** errors most harmful to stakeholders

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Which one is best or more important? It depends!

Examples

Spam Filtering



1	Key stakeholders?	user with email account; user's correspondents
2	***Type 1*** error?	user doesn't receive important email, correspondent is unable to communicate with user
3	***Type 2*** error?	user sees spam in email inbox
4	Most harmful?	***Type 1*** errors most harmful to stakeholders

optimize for **higher precision**

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Which one is best or more important? It depends!

Examples

Smart Smoke Detector



1	Key stakeholders?	user of device; members of household; neighbors
2	***Type 1*** error?	user / members of household annoyed by false alarms
3	***Type 2*** error?	user / members of household don't evacuate during fire; neighbors have less time to respond as fire spreads
4	Most harmful?	***Type 2*** errors most harmful to stakeholders

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Which one is best or more important? It depends!

Examples

Smart Smoke Detector



1	Key stakeholders?	user of device; members of household; neighbors
2	***Type 1*** error?	user / members of household annoyed by false alarms
3	***Type 2*** error?	user / members of household don't evacuate during fire; neighbors have less time to respond as fire spreads
4	Most harmful?	***Type 2*** errors most harmful to stakeholders

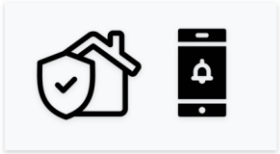
optimize for **higher recall**

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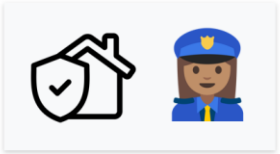
Which one is best or more important? It depends!

Examples

Home Surveillance



User Notified



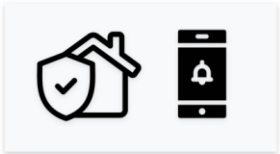
Police Notified

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Which one is best or more important? It depends!

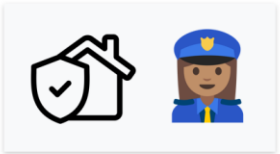
Examples

Home Surveillance



User Notified

Type 2 error is most harmful
higher recall is preferred



Police Notified

Type 1 error harms indirect stakeholders
slightly higher precision is preferred

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Some questions

Machine learning models are prone to biases in the data.

a) True

b) False

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Some questions

Which of the following are good ways to build societal trust into an artificial intelligence (AI) system? Select all that apply.

a) Efficiency

b) Lawful

c) Robust

d) Ethical

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Some questions

Machine learning is often easier to maintain and debug than traditional (deterministic) programming.

a) True

b) False

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Some questions

What is a Type 1 error?

a) A true positive

b) A false positive

c) A true negative

d) A false negative

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Some questions

What is a Type 2 error?

a) A true positive
b) A false positive
c) A true negative
d) A false negative

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Some questions

Setting a threshold that tries to minimize type 2 errors (but increases type 1 errors) has been optimized for:

a) High precision
b) High recall
c) High accuracy
d) High efficiency

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